

# Limits – 1

## Homework

1 сентября 2026 года

**Theoretical minimum.** The Cauchy criterion; monotone convergence and Bolzano–Weierstrass; Stolz–Cesàro; Fekete’s subadditive lemma; fixed points of a recursion  $x_{n+1} = f(x_n)$  and the contraction estimate; telescoping and the comparison of a sequence with an integral;  $na_n \rightarrow 0$  for a convergent series with monotone terms.

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1. (IMC 2019) Evaluate the product  $\prod_{n=3}^{\infty} \frac{(n^3 + 3n)^2}{n^6 - 64}$ .
2. (IMC 2011) Let  $(a_n)_{n \geq 0}$  satisfy  $\frac{1}{2} < a_n < 1$  for all  $n$ , and define  $x_0 = a_0$ ,  $x_{n+1} = \frac{a_{n+1} + x_n}{1 + a_{n+1}x_n}$ . What are the possible values of  $\lim_{n \rightarrow \infty} x_n$ ? Can such a sequence diverge?
3. Let  $a_1 = 2$ ,  $a_2 = 1$  and  $\frac{2}{a_n} = \frac{1}{a_{n-1}} + \frac{1}{a_{n-2}}$  for  $n \geq 3$ . Find  $\lim_{n \rightarrow \infty} a_n$ .
4. (IMC 2012) Define  $a_0 = 1$ ,  $a_1 = \frac{1}{2}$  and  $a_{n+1} = \frac{na_n^2}{1 + (n+1)a_n}$  for  $n \geq 1$ . Show that the series  $\sum_{k \geq 0} \frac{a_{k+1}}{a_k}$  converges and determine its value.
5. Prove that the sequence  $x_1 \in (0, 1)$ ,  $x_{n+1} = x_n + \frac{x_n^2}{n^2}$ , is bounded.
6. Let  $a > 0$  and let  $(x_n)_{n \geq 1}$  satisfy  $x_1 = a$  and  $x_{n+1} \geq (n+2)x_n - \sum_{k=1}^{n-1} kx_k$  for all  $n \geq 1$ . Find the limit of the sequence.
7. (IMC 2001) Let  $a_0 = \sqrt{2}$ ,  $b_0 = 2$ ,  $a_{n+1} = \sqrt{2 - \sqrt{4 - a_n^2}}$  and  $b_{n+1} = \frac{2b_n}{2 + \sqrt{4 + b_n^2}}$ .
  - a) Prove that  $(a_n)$  and  $(b_n)$  are decreasing and converge to 0.
  - b) Prove that  $(2^n a_n)$  is increasing,  $(2^n b_n)$  is decreasing and that both converge to the same limit.
  - c) Prove that  $0 < b_n - a_n < \frac{C}{8^n}$  for some positive constant  $C$ .

8. (IMC 2025 shortlist) Let  $n \geq 1$  and let reals  $a_1, \dots, a_n$  be given. For  $k > n$  define  $a_k$  as the mean of the  $n$  preceding terms. Show that  $(a_k)$  converges and determine its limit.
9. (IMC 1995) Let  $(b_n)_{n \geq 0}$  be positive reals with  $b_0 = 1$  and  $b_n = 2 + \sqrt{b_{n-1}} - 2\sqrt{1 + \sqrt{b_{n-1}}}$ . Calculate  $\sum_{n \geq 1} b_n 2^n$ .
10. (IMC 2010)
- a) A sequence satisfies  $x_{n+1} = x_n \cos x_n$  for all  $n \geq 1$ . Does it follow that the sequence converges for every initial value  $x_1$ ?
  - b) A sequence satisfies  $y_{n+1} = y_n \sin y_n$  for all  $n \geq 1$ . Does it follow that the sequence converges for every initial value  $y_1$ ?